

CHAPTER FIVE

System Design

5.1 Introduction

Chapter Four established what the Al-Khair system must do: it defined the users, expressed their needs as functional and non-functional requirements, and modeled the resulting behavior and structure through UML diagrams, an entity-relationship diagram, and a data dictionary that together fixed the database the system is built upon. This chapter addresses a different question, namely how the system presents itself to the people who use it. It covers the design of the system's interfaces: the input screens through which donors and volunteers supply data to the platform, and the output screens through which a charity administrator consumes dashboards, reports, and summary statistics. Each screen is traced back to the functional requirement it serves and to the database fields, defined in the data dictionary of Section 4.3.12, that it reads or writes, completing the transition from what the system must do to what the user actually sees.

5.2 System Interfaces

This section presents the design of the screens through which the three user roles interact with Al-Khair. Interfaces are separated into input interfaces, through which a user supplies data to the system, and output interfaces, through which the system presents information back to a user in the form of dashboards and reports. The application comprises twelve screens in total: seven core screens covering the donor, volunteer, and charity administrator workflows, and five reporting screens through which summary statistics are generated and exported. Each screen is presented as a high-fidelity mockup rather than a low-fidelity wireframe, reflecting the visual identity approved for the application, namely the navy, gold, and green color scheme and the Al-Khair mark introduced in Section 5.2.1, so that the design fixes not only the layout and the fields of each screen but also the styling in which they will ultimately be implemented in Chapter Six. As before, every element shown is traced back to the functional requirement or database field it satisfies.

5.2.1 Shared Authentication Interface

Figure 5.1 presents the sign-in screen, the single entry point shared by all three user roles and the interface through which FR1 is realized. The account-type selector lets a user declare whether they are registering as a donor or a volunteer, a distinction that determines whether the corresponding Volunteers record described in Table 4.3 is created alongside the shared Users record at registration. Charity administrators are provisioned separately by the platform owner rather than through self-registration, consistent with the CharityAdmins record in Table 4.4 having no public-facing creation path. The Al-Khair mark, rendered in gold over a navy gradient, appears above the form and recurs as a small badge on every subsequent screen so that the identity of the application remains visible throughout the user journey.

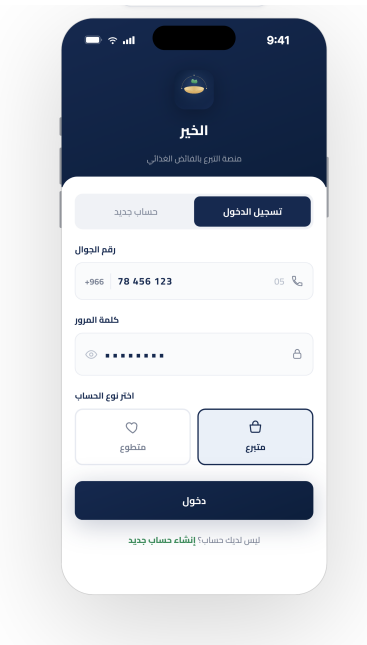


Figure 5.1. Shared interface: sign-in and account-type selection screen

5.2.2 Donor Input Interfaces

Figure 5.2 presents the donor's surplus-reporting screen, the primary input interface of the donor module. The three editable fields correspond directly to the food_category, quantity, and readiness_time attributes of the DonationReports entity in Table 4.6, and the automatically populated location field reflects FR3, under which the system captures the donor's coordinates rather than asking the donor to enter an address. The food-safety confirmation checkbox implements FR4 as a mandatory step that cannot be bypassed before the report is submitted.

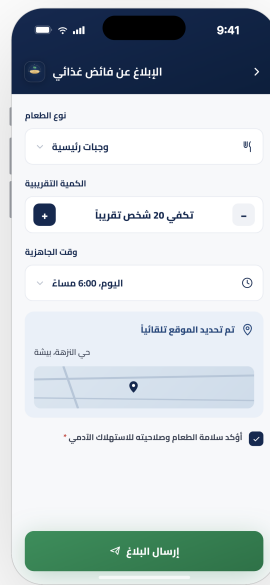


Figure 5.2. Input interface: donor surplus-reporting screen

Figure 5.3 presents the notifications and donation-status screen, which implements FR5. A donor's active donation is tracked through a horizontal progress indicator moving from collected to accepted to delivered, giving an immediate visual summary of where a report stands without requiring the donor to open the underlying record. Beneath the tracker, a reverse-chronological feed lists the individual notifications generated for that donor, each one corresponding to a record in the Notifications entity described in Table 4.7.



Figure 5.3. Output interface: donor notifications and donation-status screen

5.2.3 Volunteer Interfaces

Figure 5.4 presents the volunteer's incoming-alert screen, generated the moment a donation report matches FR6's proximity notification. The summary card draws its fields directly from the DonationReports entity, including the food category, the estimated quantity, the distance to the donor, and the time since the report was posted, and the two response buttons implement the accept-or-decline behavior specified by FR7. Accepting the alert sets the volunteer_id field on the corresponding report and triggers the navigation sequence detailed in the sequence diagram in Figure 4.8.

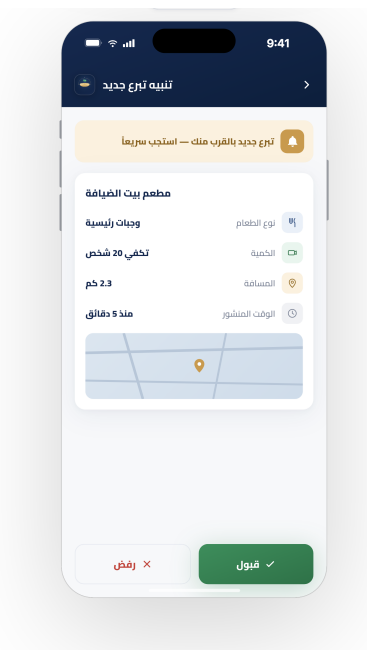


Figure 5.4. Input interface: volunteer donation-alert response screen

Figure 5.5 presents the navigation and pickup-confirmation screen through which FR8 and FR9 are carried out. A route line drawn on the map guides the volunteer from their current position to the donor's location, with the distance and estimated time shown at the top of the screen, while the two confirmation buttons at the bottom record, in sequence, the collection of the food from the donor and its subsequent delivery to the charity, each action advancing the status field of the corresponding DonationReports record.

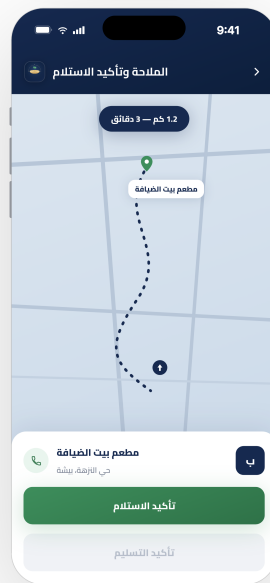


Figure 5.5. Input interface: volunteer navigation and pickup/delivery confirmation screen

5.2.4 Charity Administrator Interfaces

Figure 5.6 presents the charity administrator's dashboard, the principal output interface defined by FR11. The summary cards and the weekly donations chart give an at-a-glance view of platform activity, while the recent-donations table exposes the status field from Table 4.6 through color-coded badges, letting an administrator distinguish delivered, in-transit, and pending donations without opening individual records.

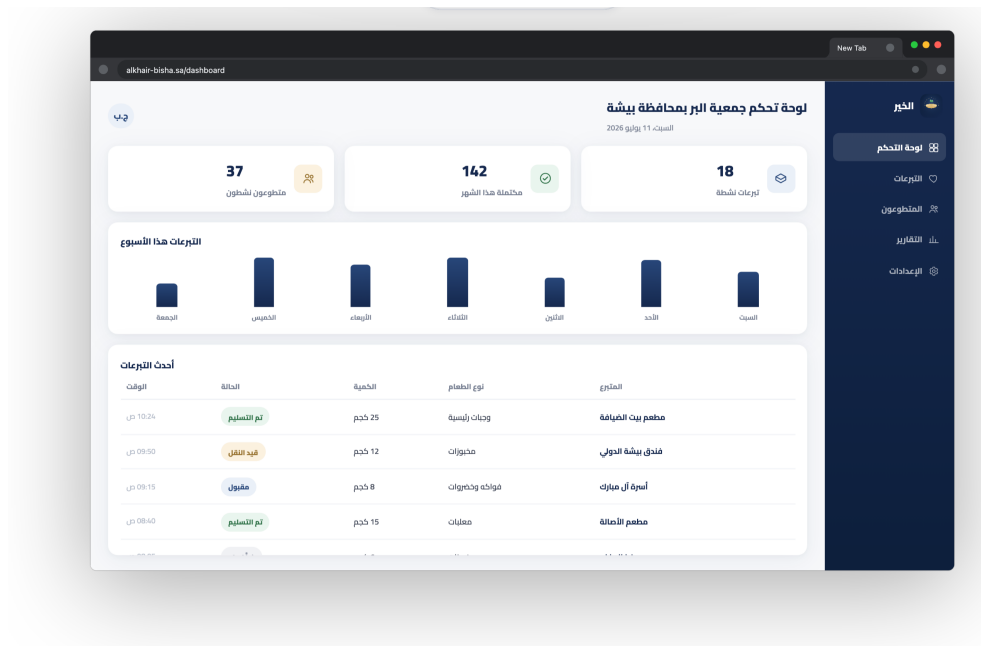


Figure 5.6. Output interface: charity administrator dashboard

Figure 5.7 presents the volunteer-approval screen through which FR10 is implemented. Each pending applicant is shown with their declared vehicle type and phone number alongside accept and reject actions, and accepting an applicant sets the approval_status field described in Table 4.3 from pending to approved, after which the volunteer becomes eligible to receive the donation alerts shown in Figure 5.4.

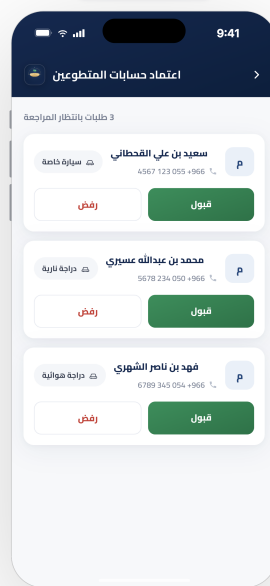


Figure 5.7. Input interface: volunteer account approval screen

5.2.5 Reporting and Export Interfaces

The remaining five screens implement FR12, under which a charity administrator generates and exports summary statistics drawn from the DonationReports entity. Figure 5.8 presents the reports landing screen, a directory of the report types available to an administrator, each entry showing when the underlying data was last refreshed.



Figure 5.8. Output interface: reports directory screen

Figure 5.9 presents the monthly summary report, which aggregates every donation record created within a selected month into a total quantity, a total donation count, and a proportional breakdown by food category shown as a pie chart. The month selector at the top allows an administrator to move between reporting periods without leaving the screen.



Figure 5.9. Output interface: monthly donation summary screen

Figure 5.10 presents the category-detail report, which restates the same monthly aggregation as a sortable table rather than a chart, giving each food category its quantity in kilograms, its donation count, and a proportional bar so that the two smaller categories are as easy to compare as the two larger ones.



Figure 5.10. Output interface: category-detail report screen

Figure 5.11 presents the volunteer-performance report, which ranks approved volunteers by the number of donations they have completed and marks the leading volunteer with a distinguishing badge, giving an administrator a basis for recognizing sustained volunteer contribution.



Figure 5.11. Output interface: volunteer-performance report screen

Figure 5.12 presents the exportable print layout that any of the preceding reports can be rendered into, carrying the Al-Khair mark, the charity name, and the generation date in its

header and the same category table demonstrated in Figure 5.10 in its body. The download and share actions at the bottom of the screen allow an administrator to hand the report to municipal or donor stakeholders as a standalone document rather than a live screen.

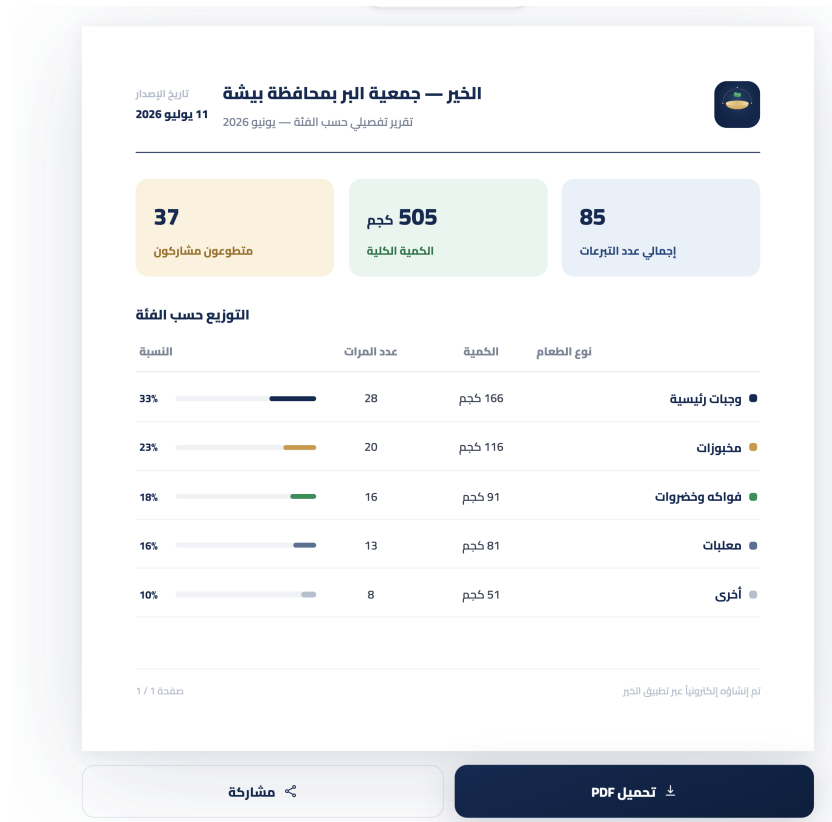


Figure 5.12. Output interface: exportable report print layout

5.3 Summary

This chapter translated the requirements and models established in Chapter Four into the concrete interface design of the Al-Khair application. Twelve screens were designed across the donor, volunteer, and charity administrator roles: the shared sign-in interface, the donor and volunteer input interfaces, the charity administrator's dashboard and approval interface, and the five reporting and export interfaces. Each screen was traced to the functional requirement it serves and to the database fields, defined in the data dictionary of Chapter Four, that it reads or writes, and each was rendered in the visual identity approved for the application rather than as a preliminary wireframe. With the interfaces now fully specified, the study proceeds to Chapter Six, which reports on the implementation, testing, and evaluation of the system built from this design.